Design and Control of One-DOF Helicopter System

Conference Paper · December 2021

2 authors:

Mehmet Akif Macit
Erzurum Teknik Üniversitesi

Ali Unluturk
Erzurum Technical University

ICAEAST'21

1st International Conference on Advances in Engineering, Architecture, Science and Technology, Erzurum Technical University, December 15-17, 2021, ERZURUM/TURKEY

Pages: 10 - 14, Paper ID: 162

Design and Control of One-DOF Helicopter System

Mehmet Akif Macit [1*][0000-0001-9450-2391] and Ali Unluturk [2][0000-0001-9722-1587]

[1]macitakif@gmail.com, Department of Electrical-Electronics Engineering, Faculty of Engineering and Architechture, Erzurum Technical University, Erzurum

[2]ali.unluturk@erzurum.edu.tr, Department of Electrical-Electronics Engineering, Faculty of Engineering and Architechture, Erzurum Technical University, Erzurum

Abstract

In this study, a One-Degree of Freedom (One-DOF) helicopter system that can move vertically up and down in the x-axis direction has been designed. A single rotor Brushless Direct Current (BLDC) motor, which is widely used in unmanned aerial vehicles, has been mounted on this One-DOF helicopter. It is aimed to enable it to fly by adding a single bladed propeller to the brushless direct current motor. There is a GY-88 10 DOF Inertial Measurement Unit (IMU) on the One-DOF helicopter system. On the GY-88 10 DOF IMU Module, it includes MPU 6050 motion sensor (3-axis gyroscope and 3-axis acceleration sensor), HMC5883L (3-axis magnetometer) module and BMP085 barometer pressure sensor on a single board for a complete orientation and heading measurement. Stable tilt angle measurement has been performed by combining 3-axis gyroscope and 3-axis acceleration sensors data with the developed Kalman Filter (KF) software. Closed-loop controller algorithms such as Proportional (P), Proportional Integral (PI) and Proportional Integral Derivative (PID) have been developed for the control of the vertical take-off and landing One-DOF helicopter system. The performances of these developed classical controllers have been evaluated in detail in the experimental results section.

Keywords: One-DOF Helicopter, BLDC Motor, PID Controller, Kalman Filter, IMU.

1. Introduction

Unmanned Aerial Vehicles (UAVs) attract the attention of the scientific community, the private sector and hobby researchers thanks to their flexible maneuvering capabilities. Today, small-sized hobby-purpose unmanned aerial vehicles, which are light, economical and capable of fast forward flight by hovering in the air in narrow spaces, are easily accessible (Bristeau et al., 2011). Also, different types of UAVs have been designed recently in order to respond to the needs of people in an efficient, economical and reliable way. For example, different flying car models continue to be developed for urban air transportation, urban passenger transportation, health sector, cargo transportation and logistics support activities in the military. The use of unmanned aerial vehicles in civilian applications, especially in the military field, has become widespread and has gained an important place. In addition to these, UAVs are the most important and practical tools for on-site response, especially in natural disasters and emergencies.

The complex dynamic model and non-linear structure of these aircraft, it makes the work of researchers very difficult (Jafar et al., 2016). Unmanned aerial vehicles such as helicopters are nonlinear systems. Therefore, it is unstable and difficult to model (Humaidi and Hasan, 2019). There is also no systematic and general adaptive controller design procedure that can be applied to all nonlinear systems such as UAVs. Therefore, the properties and operating procedures of nonlinear systems are usually converted into a linear and specific system. Then, this linear structure is analyzed and the system is controlled by using controllers such as classical P, PI, PD and PID. PID controllers are widely used because of their simple structure (Kim, 2012). The most preferred controller type in robotic applications is PID (Mane et al., 2013; Rithirun et al., 2021). Unmanned aerial vehicles also have higher system tolerance due to their usage areas. Therefore, it is necessary to match the properties and operating procedures of the nonlinear system with a linear and specific system. After the analysis of the structure, efficiency can be obtained by using PI, PD and PID controller.

In this paper, we introduce a control methodology to stabilize the One-DOF arm around the balance point, and also in which degree it is desired, without any model, and then in next attempts we will present how to control the One-DOF helicopter by P, PI and PID type controller.

The rest of this paper has been orgaized as follows. In Section II, hardware structure and software of One-DOF helicopter have been introduced. In Section III, controller performances has been experimentally tested for different scenarios. Finally, some concluding remarks have been presented in Section IV.

2. Hardware Structure and Software

Firstly, the first prototype model of the One-DOF helicopter system was obtained with SolidWorks, the computer aided 3D solid modeling and design software of which Erzurum Technical University has a license server (Erzurum Technical University, 2021). A real-time mechanical arm system that can move up and down in the x-axis direction according to the model dimensions obtained from the SolidWorks software was built. Equipment of the One-DOF helicopter system: 2200kV BLDC motor (The model number of the BLDC is 1xA2212/6T), Electronic Speed Control (ESC) unit, 3S Lithium Polymer (Li-Po) battery, GY-88 10 DOF IMU sensor board and Ardiuno UNO microprocessor. Arduino UNO is a portable embedded device that allows users to design and control robotic or mechatronic systems. The block diagram in Figure 1 shows the hardware structure of the One-DOF helicopter in detail.

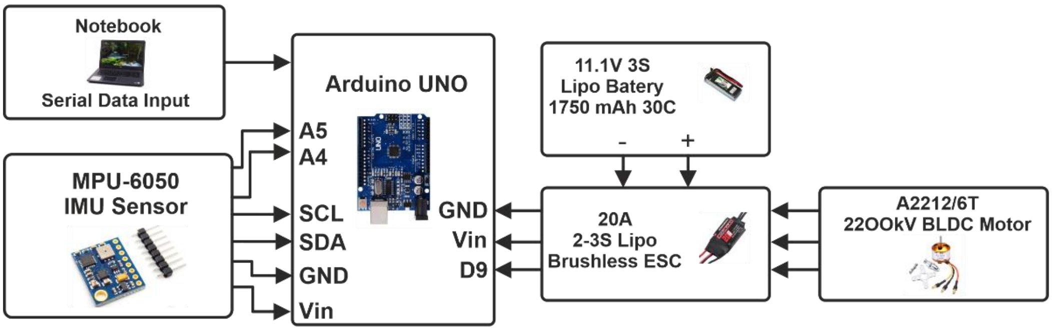

Figure 1. Hardware structure of One-DOF helicopter

There is a BLDC motor on the 1-DOF helicopter. Technical parameter of the BLDC motor is given in Table 1 (ABRA 2021).

Table 1. Parameter of the BLDC motor

MODEL	KV (rpm/V)	Voltage (V)	Prop	Load Current (A)	Power (W)	Pull (g)	Efficiency (g/W)	Li-Po Cell	Weight (g)
A2212	930	**11.1**	1060	9.8	109	660	6.1	**2-4S**	**52**
	1000		**1047**	**15.6**	**173**	**885**	**5.1**		
	1400		9050	19.0	210	910	4.3		
	1800		8060	20.8	231	805	3.5		
	2200		6030	21.5	239	732	3.1	2-3S	
	2450		6030	25.2	280	815	2.9		

As can be seen from Table 1, the BLDC motor rotates at 2200 rpm per volt. The rated voltage of the motor is 11.1V. The load current is 15.6 A, the shaft power of the motor is 173W. It has an efficiency of 5.1g/W. BLDC motor can be powered by 2-4S li-po battery.

The mechanical structure of the One-DOF helicopter system consists of a base and two vertical plates. The base plate is 180 mm wide and 400 mm long. The vertical plates have a length of 250 mm and a height of 350 mm, with a support plate between them. There is a cylindrical structure to which the main rod is connected by bearings between the two plates. The main bar has a width of 50 mm and a length of 650 mm. The elements that make up the system are placed on this bar. The SolidWorks drawing of One-DOF helicopter is shown in Figure 2a. The main parts of the realized One-DOF helicopter can be seen in Figure 2b.

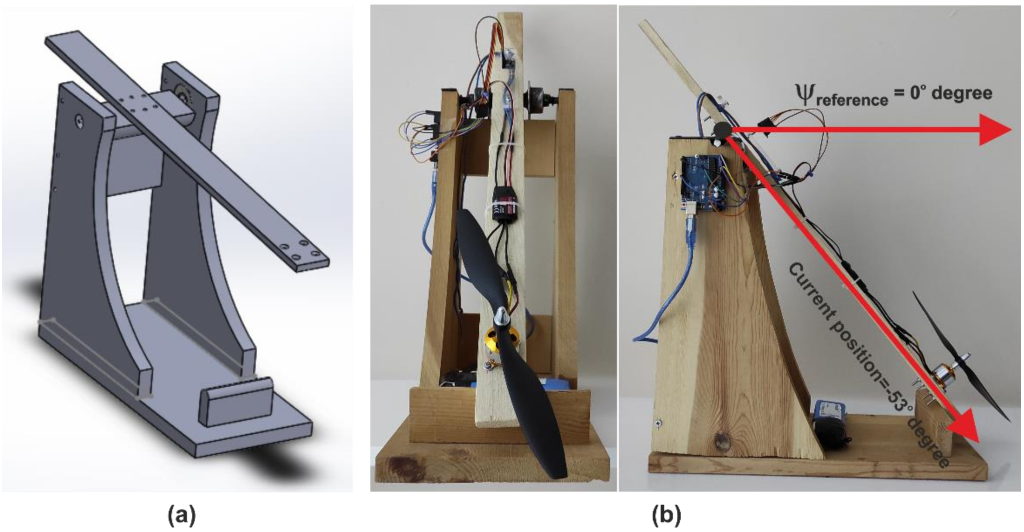

Figure 2a) Solidworks model of One-DOF helicopter **2b)** Realized One-DOF helicopter

3. Experimental Results

Classical P, PI and PID controller structures have been applied in the stability control of the One-DOF helicopter system. A low cost IMU (GY-88) sensor is used to keep the system at the desired reference point. The IMU is widely used to determine motion, orientation, and altitude. In IMU control applications, it provides information about the dynamics of the system and provides feedback to the controller. This IMU contains sensor gyroscope and acceleration sensors. The tilt angle was obtained by combining the data obtained from the gyroscope and acceleration sensor with the KF software. The tilt angle value of the One-DOF helicopter system is approximately -53° degrees when in the current position at rest. PID controller structures were tried in the stability control of the One-DOF helicopter. The realized PID controller parameter coefficients Kp, Ki and Kd were determined experimentally. The PID controller structure is given in Figure 3.

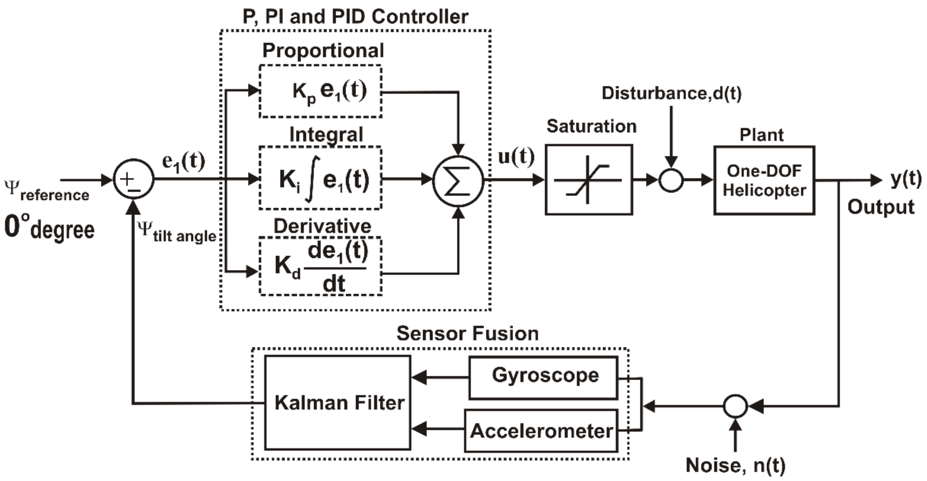

Figure 3. PID controller structure of One-DOF helicopter

The system was first checked with the proportional control method. Different proportional gain ($K_p = 0.57, K_p = 1.14, K_p = 1.71$) values applied to the system. The tilt angle change is shown in Figure 4. As can be seen in Figure 4, the reference point of the One-DOF helicopter could not be reached to 0° degrees at different proportional controller gain values. As a result, the control of the system cannot be performed only with a proportional controller.

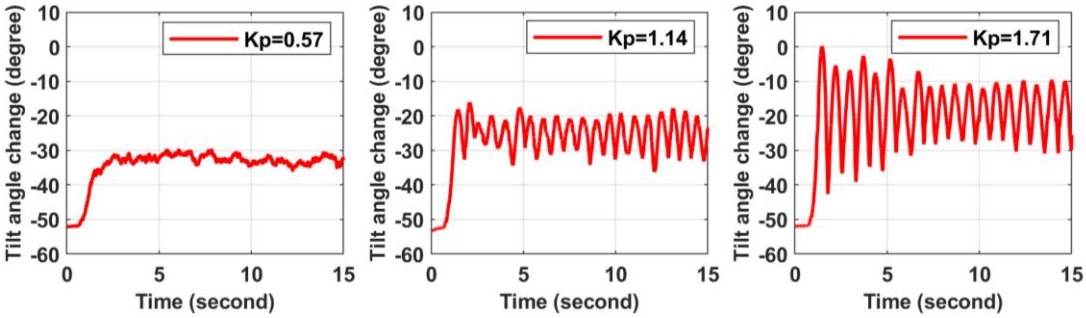

Figure 4. Different proportional controller gains of One-DOF helicopter

Secondly, proportional-integral controller was tested on the One-DOF system. Figure 5 shows the graph obtained for different PI ($K_p = 0.17\ K_i = 0.0016, K_p = 0.37\ K_i = 0.01, K_p = 0.5\ K_i = 0.0015$) controller gain values. As can be seen from the figure, the system tends to go to the reference point by excessive oscillation. Excessive oscillations are not a desirable situation in the control of systems.

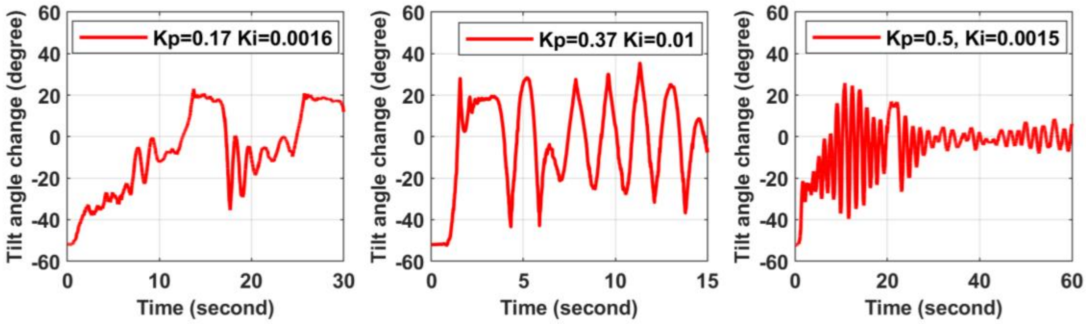

Figure 5. Different proportional-integral controller gains of One-DOF helicopter

Finally, the proportional-integral-derivative controller structure was tested on the One-DOF system. Figure 6 shows the graph obtained for different PID controller gain values. As can be seen from the figure, the system went to the reference point in a more stable way. The stable state of the system was obtained with the realized Kp= 0.57, Ki=0.0017 and Kd=0.016 PID gain parameters.

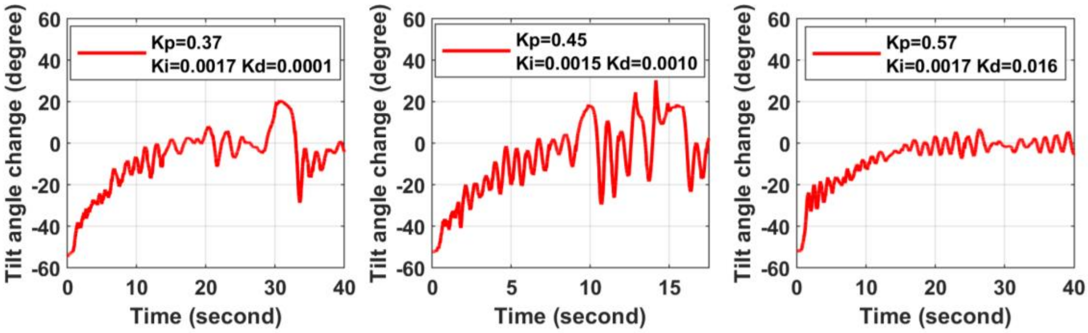

Figure 6. Different proportional-integral-derivative controller gains of One-DOF helicopter

4. Conclusion

In this study, performances of classical P, PI and PID based controllers developed for stable control of One-DOF helicopter around balance point are presented. All real-time system responses have been received and stabilization has been achieved taking into account sensor feedback. As a result of the real-time experimental results, it has been seen that the One-DOF helicopter fails the balance point control at different proportional controller gain values Kp=0.17, Kp=1.14, Kp=1.71. In the tested PI controller gain parameters (Kp= 0.17, Ki=0.0016, Kp= 0.37, Ki=0.01, Kp= 0.5, Ki=0.0015), excessive oscillations occurred in the transient state transition during the placement of the One-DOF helicopter to the reference point. In the PID controller, on the other hand, the balance stability is not good in some parameters (Kp= 0.37, Ki=0.0017, Kd=0.0001 and Kp= 0.45, Ki=0.0015 and Kd=0.0010). However, with the PID gain parameters Kp= 0.57, Ki=0.0017 and Kd=0.016, the stable state of the system was obtained. In this PID parameter, the transient and steady state response of the system is quite successful. In future studies, it is aimed to implement advanced controller structures on the system.

References

ABRA (2021). “A2212/6T 2200KV Berushless DC Motor for RC Quadcopters Planes Boats Vehicles and DIY Kits”. Web Adress: https://abra-electronics.com/quadcopters/motors/mot-qc-02.html, Date of Access: 01.12.2021.

Bristeau, P., Callou, F., Vissière, D., & Petit, N., (2011). “The Navigation and Control technology inside the AR. Drone micro UAV”. 18th IFAC world congress, 18(1), 1477-1484, Milano, Italy.

Erzurum Technical University, (2021). “Lisanslı Yazılımlar”, Web Adress: https://destek.erzurum.edu.tr /kb/faq.php?id=12, Date of Access: 03.12.2021, Erzurum, Turkey.

Humaidi, A. J., & Hasan, A. F., (2019). “Particle swarm optimization–based adaptive super-twisting sliding mode control design for 2-degree-of-freedom helicopter”. Measurement and Control, Vol. 52(9-10), 1403-1419.

Jafar, A., Ahmad, S. M. & Ahmed N. (2016). "Mathematical modeling and control law design for 1DOF Quadcopter flight dynamics," 2016 International Conference on Computing, Electronic and Electrical Engineering (ICE Cube), pp. 79-84, IEEE, Quetta, Pakistan.

Kim, D. H., (2012). “Design and tuning approach of 3-DOF emotion intelligent PID (3-DOF-PID) controller”. In 2012 Sixth UKSim/AMSS European Symposium on Computer Modeling and Simulation, pp. 74-77, IEEE. Malta.

Mane, R., Mahapatro, K. A. & Gundecha, A. D., (2019). “Single Degree of Freedom Helicopter Model: Laboratory Setup Design”. 2019 International Conference on Advances in Computing, Communication and Control (ICAC3), IEEE, Mumbai, India.

Rithirun, C., Charean, A., & Sawaengsinkasikit, W. (2021). “Comparison Between PID Control and Fuzzy PID Control on Invert Pendulum System”. In 2021 9th International Electrical Engineering Congress (iEECON), pp. 337-340, IEEE. Pattaya, Thailand.